

AI & DATA SCIENCE CAREER ROADMAP

B.Tech + M.Tech CSE • Integrated 5-Year Program • India • 2025–2030

AI Engineer • Data Scientist • SSC CHSL/CGL • 5-Year Plan

CURRENT POSITION & GOALS	SEMESTER-WISE MILESTONES	SUGGESTED DAILY SCHEDULE
Student: Integrated B.Tech+M.Tech CSE Level: Complete Beginner (Zero Coding) Goals: AI Engineer Data Scientist SSC Timeline: 5yr SSC CHSL • 5yr Tech Career	SEM 1 Python syntax, loops, functions, Git basics SEM 2 OOP, File I/O, 1st mini-project, LinkedIn setup SEM 3 DSA (Arrays→Linked List), SQL, Kaggle start SEM 4 DSA Advanced, ML basics, 1st internship SEM 5 Deep Learning, NLP, portfolio projects SEM 6 CV/NLP projects, research, 2nd internship SEM 7 MLOps, system design, open source SEM 8 MTech research begins, publications SEM 9 Thesis + job prep + final internship SEM 10 Placement / Full-time offers secured!	5:30–6:00 Morning routine + planning 6:00–7:00 SSC Prep (Quant / English) 7:00–7:30 Breakfast + commute 8:00–2:00 College classes (attend fully!) 2:30–4:00 Coding / DSA (LeetCode/HackerRank) 4:00–5:00 AI/ML / Online Course (Coursera) 5:00–5:30 LinkedIn — 1 post or connection 5:30–6:30 Project work / assignments 6:30–8:00 Dinner + relaxation + revision 8:00–9:00 SSC Mock test or PYQ practice
STRENGTHS & SKILL GAPS	LINKEDIN PERSONAL BRANDING	
STRENGTHS - Early start → 5 years to master AI - Integrated MTech = research advantage - Young & adaptable mindset - Dual track: Tech + Govt backup SKILL GAPS - No coding/programming basics	Month 1: Complete profile: photo, banner, headline Month 2: Post: 'I'm learning Python' journey posts Month 3: Share 1st project with GitHub link Month 4: Connect 500+ (professors, developers) Ongoing: 2-3 posts/week: learnings, projects, tips 6 Months: Target: 1000+ followers, 50+ connections	
LEARNING ROADMAP (5 YEARS)	PROJECT ROADMAP	DSA PREPARATION STRATEGY
Yr 1 Python Basics → OOP → Git → DSA Foundations Yr 2 DSA Advanced → SQL → ML Basics → Flask Yr 3 Deep Learning → NLP → Computer Vision → APIs Yr 4 MLOps → System Design → Research Papers Yr 5 MTech Research → Publications → Dream Job	BEGINNER (Yr 1-2) - Calculator / Number Guessing Game - Student Grade Tracker (Python) - Weather App (API integration) INTERMEDIATE (Yr 2-3) - Movie Recommendation System - Sentiment Analysis (Twitter/IMDb) - Image Classifier (CNN - CIFAR-10) ADVANCED (Yr 3-5) - AI Chatbot with RAG (LangChain) - Real-time Object Detection App - End-to-End ML Pipeline (MLOps)	Phase 1 (Yr 1) Arrays, Strings, Basic Recursion Phase 2 (Yr 2) LL, Trees, Stacks, Queues, Hashing Phase 3 (Yr 3) Graphs, DP, Greedy, Binary Search Platform: LeetCode Easy→Med • Striver's A-Z Target: 150+ problems solved by Yr 2 end
SSC CHSL & CGL STRATEGY	INTERNSHIP & JOB STRATEGY	IMMEDIATE 30-DAY ACTION PLAN
SSC CHSL (1 YEAR): Target: 2025-26 Exam SSC CGL (4 YEARS): Backup: 2029 Exam Daily SSC Schedule (1hr/day): 6:00–6:30 AM → Quantitative Aptitude 6:30–7:00 AM → English / Reasoning - Weekend → Mock Tests + PYQs - Apps: Oliveboard, Testbook, Adda247	Sem 3-4: Apply: InternPe, Internshala, LetsGrowMore Sem 4-5: Data Analyst intern → build real projects Sem 6-7: SDE/ML intern at product startup Sem 8-9: Research intern or Pre-Placement Offer Yr 5: Full-time: █12-25 LPA target (AI/ML roles)	Day 1-3 Install Python + VS Code + GitHub acc Day 4-7 Python: variables, loops, functions Day 8-10 Create LinkedIn profile (complete it) Day 11-15 Python: lists, dicts, OOP basics Day 16-18 Build: Number Guessing Game project Day 19-22 Start SSC CHSL Quant (Basic Arithmetic) Day 23-25 Push project to GitHub + share LinkedIn Day 26-28 LeetCode: solve 5 easy problems Day 29-30 Review progress + set Month 2 targets